

REM Part 3: Ensuring Comprehensive Data Coverage and Parsimony



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In this third and final blog post to introduce the Mina Real Estate Modeling (REM) framework, I want to go over some of the technical aspects of what makes our modeling approach reliable and what makes our dataset unique and comprehensive. As in our prior two blog posts about Mina's REM framework, our goal in sharing information about our framework with prospective clients and investors is to allow them to see a bit inside the black box of our statistical modeling, and share our enthusiasm and confidence in the REM framework we have built.

First, here is a review of the six major discoveries revealed by our REM framework, which were explained in our first two blog posts.

1. Green certification – that is, either an Energy Star Certification or a LEED Certification – has a large average positive consequence for revenue per square foot in office buildings in Florida and California, which reflects the willingness of tenants, on average, to pay more for space that is green-certified.
2. Green-certification effects, however, are not the same across all locations within a state. Which attributes are reliably important for influencing the size of the green-certification premium varies across

the existence of green premia, and their certification choices reveal implicit opinions about how green certification premia vary across locations.

4. Market participants' implied opinions (shown in Table 1) about which local population attributes are associated with greater rewards for green certification are directionally correct, but they are incomplete. There is substantial money left on the table from sub-optimal green certification choices. The disciplined statistical analysis of the REM framework is able to guide certification decisions to identify the low-hanging fruit that market participants so far have missed: buildings that are currently not green-certified, but that would experience large increases in leasing revenues from becoming green-certified.

5. Including a comprehensive list of building attributes in the REM framework is essential for avoiding an overstatement of the consequences of green certification. For example, new buildings are more likely to be green-certified. Because age has a negative effect on lease revenue per square foot, if one does not control for the age of the office building, one will attribute too much of an effect to green certification.

6. Building attributes (amenities, size, etc.) matter within a given location for determining the size of the green premium. This interaction between green certification and building attributes can be very large, and is an important factor to include in modeling. In one example from our second blog post we showed that, within the same zip code, a combination of attributes implied a green premium of 7% for one office building and 13% for another.

In this blog post, I explain how the REM framework ensures that our coverage of building and locational attributes is comprehensive, on the one hand, and not excessive, on the other hand. These two concerns are the Sylla and Charybdis of statistical modeling: either leaving out important data ("omitted variable bias") or including spurious variables in a model ("overfitting") can make a model's estimates of the revenue consequences of a building characteristic – such as green certification, or a building amenity – unreliable.

The good news, however, is that there are tried and true methods of "validation" to avoiding the overfitting problem, and new methodologies for expanding the range of variables that a model can consider by making use of artificial intelligence (AI) and natural language processing (NLP). The REM framework includes all of those tools. While no modeling framework can ever be perfect, and we will continue to hone our craft in improving our models over time, we are confident that the REM framework is employing state-of-the-art methods for making the data inclusion choices of our models both comprehensive and properly parsimonious.

A key problem with omitted variables bias is that if you don't have an important variable in the model, that will generally bias the estimated effects of the variables you do include in the model, if those included variables are correlated with the missing one. In real estate, everyone knows that "soft" information can be hugely important for the lease revenue: How attractive is the design of the building? How accessible is the location? How nice is the view? Those attributes vary both within and across zip codes. How can one incorporate such considerations about building characteristics alongside other "hard" data (such as building size, age, and the presence of various amenities)?

The proprietary database we have constructed over the past three years includes many measures of soft information that address these and other hard-to-measure aspects of office properties. The key to including those variables is finding ways to quantify textual information. We collect textual information

AI is a crucial tool for developing routinized procedures for reliably reading and coding textual information. NLP is a toolkit for distilling characteristics of textual information that capture nuances of the communication, such as the positivity or negativity of its tone, the novelty of the word usage, or the extent to which the text gives weight to a topical area. By combining AI and NLP methods in our data collection efforts, we ensure a wide range of soft-information indicators from which one can choose when selecting which data to include in any statistical model. That is what we mean by a comprehensive database.

Overcoming the problem of overfitting – including too much data in a model, some of which may produce spurious estimates – is addressed using two traditional modeling tools: forward selection and cross-validation. Forward selection is a way to select variables to use in a model from a long list of potential variables that might be included. We employ more than one criterion for variable selection to ensure that all potentially useful variables are included. Part of the art of modeling is deciding on selection criteria that are strict enough but not too strict.

Table 1: Percentage of Office Buildings That Are Green Certified, By Population Types

Florida	Energy Star-Certified	LEED-Certified
% in Highest Democrat voting quartile	6.2%	4.9%
% in Highest education quartile	7.6%	7.7%
% in Highest income quartile	7.0%	7.3%
% in Lowest Democrat voting quartile	0.0%	0.0%
% in Lowest education quartile	2.0%	2.2%
% in Lowest income quartile	2.5%	2.1%
Total % of office buildings	5.0%	4.1%

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California	Energy Star-Certified	LEED-Certified
% in Highest Democrat voting quartile	13.8%	10.7%

% in Highest education quartile	15.7%	10.3%
% in Highest income quartile	13.6%	10.2%
% in Lowest Democrat voting quartile	5.7%	0.9%
% in Lowest education quartile	5.1%	1.4%
% in Lowest income quartile	6.8%	3.6%
Total % of office buildings	11%	7.2%

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